Lab 4: The Simple Pendulum

Each group timed twenty swings of a pendulum for four string lengths and computed the gravitational acceleration from the period. The period of a simple pendulum for small angles is

$$T = 2\pi\sqrt{\frac{L}{g}}, \qquad g = \frac{4\pi^2 L}{T^2}. \tag{1}$$

Table 1 lists the measured values.

Table 1: Measured periods and the derived value of g.

Trial	Length (m)	Period (s)	g (m s^{-2})
1	0.250	1.004	9.79
2	0.500	1.420	9.79
3	0.750	1.737	9.81
4	1.000	2.006	9.81

The values agree within one per cent. The uncertainty in each period comes mostly from the reaction time of the person holding the stopwatch, which is about 0.1 s for a single swing but is divided by twenty when twenty swings are timed together.

Thermal Properties of Common Materials

Table 2 groups the materials by type. The specific heat is given for the solid and the liquid phase where both are common in the laboratory.

Table 2: Specific heat and latent heat of fusion.

Type	Material	Specific heat (J kg^{-1} K^{-1})		L_f (kJ kg^{-1})
		Solid	Liquid	
Metal	Aluminium	900	1180	397
	Copper	385	530	205
	Lead	128	140	24.5
Other	Water	2090	4186	334
	Ethanol	2420	2440	108

Summary of Kinematics and Dynamics

The quantities below appear in every chapter of the course. Each definition is written with the symbols used in the lectures.

Table 3: Definitions of mechanical quantities.

Quantity	Symbol	Definition	SI unit
Average velocity	$\bar{v}$	$\bar{v} = \dfrac{\Delta x}{\Delta t}$	$\mathrm{m\,s^{-1}}$
Acceleration	a	$a = \dfrac{dv}{dt}$	$\mathrm{m\,s^{-2}}$
Momentum	$\vec{p}$	$\vec{p} = m\vec{v}$	$\mathrm{kg\,m\,s^{-1}}$
Kinetic energy	K	$K = \tfrac{1}{2}mv^2$	J
Work	W	$W = \int \vec{F} \cdot d\vec{r}$	J

A quick check of units is the fastest way to catch an algebra mistake: the kinetic energy has units of $\mathrm{kg\,(m\,s^{-1})^2}$, which is a joule.

Table 4: Frequency of quiz scores.

Score	Frequency	Relative frequency	Cumulative relative frequency
5	2	0.10	0.10
6	3	0.15	0.25
7	6	0.30	0.55
8	5	0.25	0.80
9	3	0.15	0.95
10	1	0.05	1.00

Derivatives by Finite Differences

The table below gives values of $f(x) = x^3 - 2x$ and its derivative at evenly spaced points. It is set without any rules, as many instructors type tables in a hurry.

Table 5: Values of f and f'.

x	$f(x)$	$f'(x)$	Forward difference
-2	-4	10	5.0
-1	1	1	-1.0
0	0	-2	-1.0
1	-1	1	5.0
2	4	10	17.0

The forward difference with step $h = 1$ overestimates the derivative where the graph curves upward, because

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2} f''(\xi). \tag{2}$$

Table 6: Surface gravity of the planets.

Planet	**Mass (10^{24} kg)**	**Radius (km)**	g **(m/s^2)**
Mercury	0.330	2440	3.7
Venus	4.87	6052	8.9
Earth	5.97	6371	9.8
Mars	0.642	3390	3.7